



9/1/2026

N + - Network Fundamentals & Building Networks.

Network Review & Final Practice.

TASK 1.

- ❖ **Open Cisco Packet Tracer and create a new project file named 'HomeWiFiNetwork.pkt' with three PCs and one wireless router connected appropriately.**

- **Home Wi-Fi Network:**

Project File: HomeWiFiNetwork.pkt

Steps:

- Open **Cisco Packet Tracer** and create a new project.
- Add **three PCs** and **one Wireless Router** to the workspace.
- Connect the three PCs to the wireless router appropriately.
- Save the project as *HomeWiFiNetwork.pkt*.

Result: A home Wi-Fi network with **3 PCs connected to 1 wireless router** was created and saved successfully.

TASK 2.

- ❖ **Assign static IP addresses to each PC in your Packet Tracer network so they are on the same subnet (e.g., 192.168.10.x), and configure the default gateway to the router's IP.**

- **Assign Static IP Addresses:**

The three PCs were assigned **static IP addresses** from the same subnet, 192.168.10.0/24. The router's IP address was set as the **default gateway**.

Device	IP Address	Subnet Mask	Default Gateway
PC1	192.168.10.2	255.255.255.0	192.168.10.1
PC2	192.168.10.3	255.255.255.0	192.168.10.1
PC3	192.168.10.4	255.255.255.0	192.168.10.1

Result: All three PCs are on the same subnet and use the router's IP address 192.168.10.1 as their default gateway.

TASK 3.

❖ **Test connectivity by using the Packet Tracer 'ping' tool to verify that all PCs can communicate with each other and the router, Use the Command Prompt tool on each PC and type 'ping [other PC's IP]'.**

➤ **Test Connectivity Using Ping:**

Procedure:

I used the **Command Prompt** tool on each PC in Packet Tracer and tested connectivity by entering:

ping [Other PC's IP Address]

I also pinged the **router's IP address** from each PC.

Result:

The ping replies were received successfully, which confirms that **all PCs can communicate with each other and with the router.**

TASK 4.

- ❖ Document each step you took to build and configure your Packet Tracer network, including screenshots of your topology and the IP configuration windows for each PC, in a Word or PDF file.

➤ Packet Tracer Network Documentation:

Step 1: Create the Network

Opened **Cisco Packet Tracer** and created a new project.

Step 2: Add and Connect Devices

Added **two PCs, one 2960 Switch, and one Router**. Connected both PCs to the switch using Ethernet cables and connected the switch to the router.

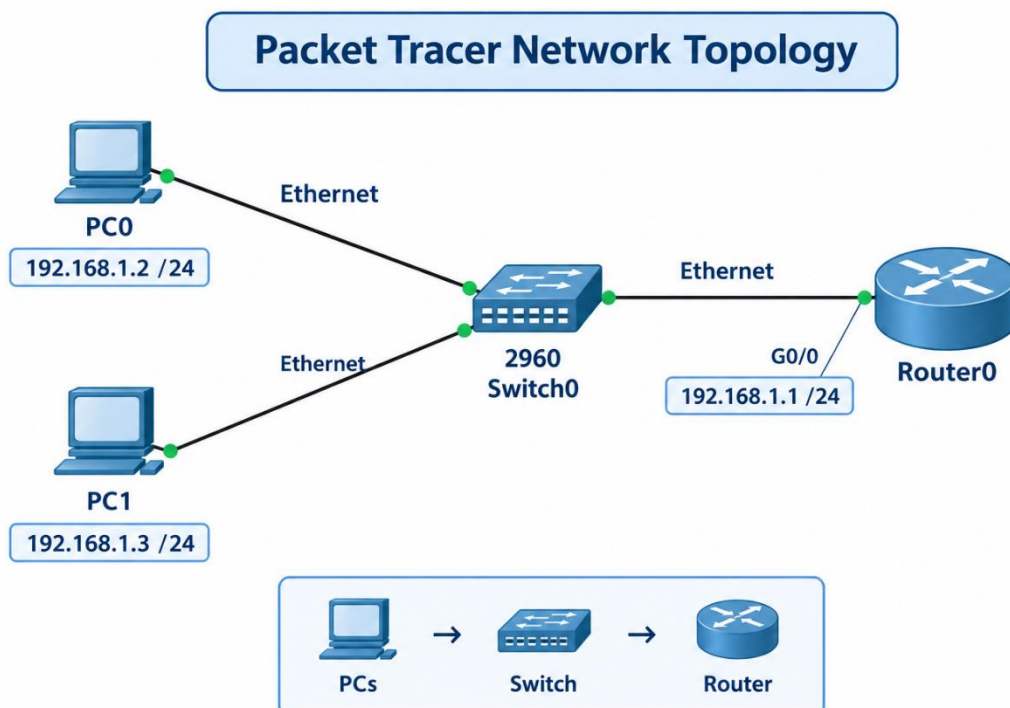


Figure : Packet Tracer Network Topology.

Step 3: Configure IP Addresses

Configured the network using the 192.168.1.0/24 subnet.

Device	IP Address	Subnet Mask	Default Gateway
Router0 (G0/0)	192.168.1.1	255.255.255.0	—
PC0	192.168.1.2	255.255.255.0	192.168.1.1
PC1	192.168.1.3	255.255.255.0	192.168.1.1

Result:

The network was successfully built and configured with the required IP addresses.